

Matematica - Clasa a 10-a

Lectii esentiale - Toate capitolele

1. Functii - aprofundare

- Monotonia: crescatoare / descrescatoare pe un interval
- Functia crescatoare: $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$
- Functia inversa f^{-1} exista daca f e bijectiva

Graficul lui f^{-1} = simetricul graficului lui f fata de $y=x$

2. Functia logaritmica si exponentiala

Functia exponentiala $f(x) = a^x$ ($a > 0, a \neq 1$)

- $a > 1$: crescatoare; $0 < a < 1$: descrescatoare
- Grafic trece prin $(0, 1)$ si $(1, a)$

$$a^x = a^y \Leftrightarrow x = y$$

Functia logaritmica $f(x) = \log_a(x)$

- Domeniu: $x > 0$; Grafic trece prin $(1, 0)$ si $(a, 1)$
- $a > 1$: crescatoare; $0 < a < 1$: descrescatoare

$$\log_a(xy) = \log_a(x) + \log_a(y)$$

$$\log_a(x/y) = \log_a(x) - \log_a(y)$$

$$\log_a(x^p) = p \log_a(x)$$

$$\text{Schimbare baza: } \log_a(x) = \log_b(x) / \log_b(a)$$

$$\log_a(a) = 1; \log_a(1) = 0; a^{(\log_a(x))} = x$$

3. Ecuatii si inecuatii exponential-logaritmice

- Ec. exponentiala: $a^{f(x)} = a^{g(x)} \Leftrightarrow f(x) = g(x)$
- Ec. logaritmica: $\log_a(f(x)) = \log_a(g(x)) \Leftrightarrow f(x) = g(x), f(x) > 0$
- Inecuatii: $a > 1 \Rightarrow$ pastreaza sensul; $0 < a < 1 \Rightarrow$ inverseaza sensul

Substitutia $t = a^x$ reduce ec. exponentiala la ec. de gr.II

4. Trigonometrie avansata

Functii trigonometrice pe cercul trigonometric:

- $\sin x$: $T=2\pi$, domeniu $\mathbb{R}$, codomeniu $[-1, 1]$
- $\cos x$: $T=2\pi$, domeniu $\mathbb{R}$, codomeniu $[-1, 1]$
- $\operatorname{tg} x$: $T=\pi$, domeniu $\mathbb{R}$ fara $\{\pi/2+k\pi\}$
- $\operatorname{ctg} x$: $T=\pi$, domeniu $\mathbb{R}$ fara $\{k\pi\}$

Formule importante:

$$\sin^2 x + \cos^2 x = 1$$

$$\sin(x+\pi) = -\sin x; \quad \cos(x+\pi) = -\cos x$$

$$\sin(\pi-x) = \sin x; \quad \cos(\pi-x) = -\cos x$$

$$\sin(-x) = -\sin x; \quad \cos(-x) = \cos x$$

$$\sin(a+b) = \sin a \cos b + \cos a \sin b$$

$$\cos(a+b) = \cos a \cos b - \sin a \sin b$$

$$\sin 2a = 2 \sin a \cos a$$

$$\cos 2a = 2\cos^2 a - 1 = 1 - 2\sin^2 a = \cos^2 a - \sin^2 a$$

Ecuatii trigonometrice de baza:

$$\sin x = a \Rightarrow x = \arcsin(a) + 2k\pi \text{ sau } x = \pi - \arcsin(a) + 2k\pi$$

$$\cos x = a \Rightarrow x = \pm \arccos(a) + 2k\pi$$

$$\operatorname{tg} x = a \Rightarrow x = \operatorname{arctg}(a) + k\pi$$

5. Vectori si sisteme de coordonate

$$v = (v_x, v_y); \quad |v| = \sqrt{v_x^2 + v_y^2}$$

$$u \cdot v = u_x v_x + u_y v_y = |u||v| \cos(\text{unghi})$$

$$u \text{ perp. } v \Leftrightarrow u \cdot v = 0$$

$$\text{Distanța } AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Mijlocul } AB: M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

6. Geometrie analitica

Dreapta in plan:

Ecuatia: $y = mx + n$ sau $ax + by + c = 0$

Panta: $m = (y_2 - y_1) / (x_2 - x_1) = \text{tg}(\text{unghi cu } Ox)$

Prin 2 pct: $(y - y_1) / (y_2 - y_1) = (x - x_1) / (x_2 - x_1)$

$d(P_0, \text{dreapta } ax+by+c=0) = |ax_0+by_0+c| / \sqrt{a^2+b^2}$

- Paralele: $m_1 = m_2$; Perpendiculare: $m_1 \times m_2 = -1$

Cercul:

$(x-a)^2 + (y-b)^2 = r^2$ (centru (a,b) , raza r)

Forma generala: $x^2 + y^2 + Dx + Ey + F = 0$

Centrul: $(-D/2, -E/2)$; Raza: $r = \sqrt{D^2/4 + E^2/4 - F}$

- Tangenta la cerc in pct $M_0(x_0, y_0)$: $(x_0 - a)(x - a) + (y_0 - b)(y - b) = r^2$

7. Matrice si sisteme liniare

Matrice 2×2 : $\det(A) = ad - bc$; inversa: $A^{-1} = (1/\det) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Regula Cramer: $x_i = \det(A_i) / \det(A)$

Gauss: aducem matricea extinsa la forma treapta

- $\det(A) = 0 \Leftrightarrow$ sistemul nu are solutie unica

8. Numere complexe

$z = a + bi$, unde $i^2 = -1$

$|z| = \sqrt{a^2 + b^2}$ (modulul); conjugat: $\bar{z} = a - bi$

Forma trigonometrica: $z = r(\cos t + i \sin t)$, $r = |z|$

Inmultire: $|z_1 z_2| = |z_1| |z_2|$; $\arg(z_1 z_2) = \arg z_1 + \arg z_2$

Formula lui Moivre: $z^n = r^n(\cos nt + i \sin nt)$

- Radacini ale ecuatiei $z^2 + 1 = 0$: $z = +i$ si $z = -i$
- Ec. $ax^2 + bx + c = 0$ cu $\Delta < 0$ are rad. complexe conjugate

Clasa 10 - Calcul diferential (intro)

9. Limite de functii - notiuni

$\lim_{x \rightarrow a} f(x) = L$ (functia tinde la L cand x se apropie de a)

$\lim_{x \rightarrow 0} \sin x / x = 1$ (limita fundamentala)

$\lim_{x \rightarrow \infty} (1 + 1/x)^x = e$ (numarul lui Euler)

Reguli: $\lim(f+g) = \lim f + \lim g$; $\lim(fg) = \lim f \times \lim g$

$\lim(f/g) = \lim f / \lim g$ (daca $\lim g \neq 0$)

- Forme nedeterminate: $0/0, \infty/\infty \Rightarrow$ L'Hopital sau simplificare

10. Continuitatea functiei

f continua in x_0 daca $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

- Functii continue: polinoame, trigo, exp, log (pe domeniu)
- Punct de discontinuitate: limita nu exista sau $\neq f(x_0)$

11. Derivata - introducere

$f'(x_0) = \lim_{h \rightarrow 0} (f(x_0+h) - f(x_0))/h$ (derivata in x_0)

Interpretare geometrica: panta tangentei la grafic in x_0

$(x^n)' = n x^{n-1}$; $(e^x)' = e^x$; $(\ln x)' = 1/x$

$(\sin x)' = \cos x$; $(\cos x)' = -\sin x$

$(f+g)' = f' + g'$; $(fg)' = f'g + fg'$; $(f/g)' = (f'g - fg')/g^2$

Compusa: $(f(g(x)))' = f'(g(x)) \times g'(x)$

12. Aplicatii ale derivatei

- $f'(x) > 0$ pe $I \Rightarrow f$ crescatoare pe I
- $f'(x) < 0$ pe $I \Rightarrow f$ descrescatoare pe I
- $f'(x_0) = 0$ si semnul schimbat $\Rightarrow$ extrem local

Ecuatia tangentei: $y - f(x_0) = f'(x_0)(x - x_0)$

Clasa 10 - Probabilitati si Statistica

13. Probabilitati conditionata si independenta

$$P(A/B) = P(A \text{ cap } B) / P(B) \text{ (prob. lui A stiind B)}$$

$$\text{Independenta: } P(A \text{ cap } B) = P(A) \times P(B)$$

$$\text{Formula probabilitatii totale: } P(A) = \sum P(A/B_i)P(B_i)$$

$$\text{Formula Bayes: } P(B_i/A) = P(A/B_i)P(B_i) / P(A)$$

14. Variabile aleatoare discrete

X v.a. discreta: ia valori $x_1, x_2, \dots, x_n$ cu prob. $p_1, p_2, \dots, p_n$

$$\text{Conditie: } p_1 + p_2 + \dots + p_n = 1$$

$$\text{Speranta matematica: } E(X) = \sum x_i p_i$$

$$\text{Dispersia: } D(X) = E(X^2) - (E(X))^2$$

$$\text{Distributia binomiala: } P(X=k) = C(n, k) p^k (1-p)^{(n-k)}$$

$$E(X) = np; D(X) = np(1-p) \text{ pt distributia binomiala}$$

15. Statistica

$$\text{Media aritmetica: } \bar{x} = \sum (x_i f_i) / n$$

$$\text{Dispersia: } s^2 = \sum ((x_i - \bar{x})^2 f_i) / n$$

$$\text{Abaterea standard: } s = \sqrt{s^2}$$

$$\text{Coeficientul de variatie: } CV = s / \bar{x} \times 100\%$$

- $CV < 30\%$: date omogene; $CV > 30\%$: date eterogene

Clasa 10 - Exemple rezolvate

Exemple pas cu pas

Ecuatie logaritmica: $\log_2(x) + \log_2(x-2) = 3$

$$\log_2(x(x-2)) = 3 \Rightarrow x(x-2) = 2^3 = 8$$

$$x^2 - 2x - 8 = 0 \Rightarrow \Delta = 4 + 32 = 36 \Rightarrow x_1 = 4, x_2 = -2$$

$x_2 = -2$ inacceptabil (log nedefinit) {ARR} $x = 4$ {OK}

Ecuatie exponentiala: $4^x - 5 \times 2^x + 4 = 0$

Substitutie: $t = 2^x$ ($t > 0$)

$$t^2 - 5t + 4 = 0 \text{ {ARR} } t_1 = 4, t_2 = 1$$

$$2^x = 4 \text{ {ARR} } x = 2; \quad 2^x = 1 \text{ {ARR} } x = 0$$

Derivata: $f(x) = x^3 - 3x$

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 3(x-1)(x+1)$$

$$f' = 0 \text{ pt } x = 1 \text{ si } x = -1$$

Crescatoare pe $(-\infty, -1)$ si $(1, +\infty)$; Descrescatoare pe $(-1, 1)$

Numar complex: $(2+3i)(1-i)$

$$= 2 - 2i + 3i - 3i^2 = 2 + i - 3(-1) = 5 + i$$

Clasa 10 - Rezumat

Log & Exp

- $\log_a(xy) = \log_a x + \log_a y$
- $a^f = a^g \Leftrightarrow f = g$
- Schimbare baza: $\log_b x / \log_b a$

Trigo avansata

- $\sin^2 x + \cos^2 x = 1$
- $\sin(a+b) = \sin a \cos b + \cos a \sin b$
- $\sin 2a = 2 \sin a \cos a$

Geometrie analitica

- Dreapta: $y = mx + n$ sau $ax + by + c = 0$
- Cerc: $(x-a)^2 + (y-b)^2 = r^2$
- $d(P, d) = |ax_0 + by_0 + c| / \sqrt{a^2 + b^2}$

Numere complexe

- $z = a + bi$, $i^2 = -1$
- $|z| = \sqrt{a^2 + b^2}$
- Moivre: $z^n = r^n (\cos nt + i \sin nt)$

Derivate

- $(x^n)' = nx^{n-1}$; $(e^x)' = e^x$
- $(fg)' = f'g + fg'$; $(f/g)' = (f'g - fg')/g^2$
- $f' > 0 \Rightarrow$ crescatoare

Probabilitati

- $P(A/B) = P(A \cap B) / P(B)$
- $E(X) = \sum x_i p_i$
- Binomiala: $E = np$, $D = np(1-p)$